

# Grounded RAG Pipeline for Code-Switching (Hinglish) in Clinical Decision Support Systems.

## ABSTRACT

Clinical AI systems in India often fail when handling naturally spoken Hinglish, leading to misinterpretations and clinically unsafe, hallucinated outputs. This arises because such inputs are treated as malformed English rather than valid code-switched language.

- We propose a **RAG-based pipeline** to ground Hinglish queries in real clinical evidence.
- Built using **LaBSE** cross-lingual embeddings and **FAISS** retrieval.
- Evaluated on **61,316** MultiCaRe clinical cases.
- Demonstrates **+73.5% improvement** in factual support and **-44.0% reduction** in hallucinations over a zero-shot baseline.

## RESEARCH GAP

A critical mismatch exists between patient language and AI model assumptions in Indian healthcare settings. Patients express symptoms in natural Hinglish, whereas models are primarily designed for formal English inputs. This reflects not a translation issue but a **reasoning alignment challenge**.

Existing approaches show **key limitations**:

- Tend to **hallucinate diagnoses** without supporting evidence
- Fail to retrieve relevant** clinical context
- Result in outputs with **no factual grounding** (Factual: 0.00, Hallucination: 1.00)

## METHODOLOGY

- Data:** 3,015 Hinglish queries (MMCQSD dataset) matched to 61,316 MultiCaRe clinical cases across 18 condition groups.
- Retrieval:** LaBSE encodes queries and evidence into a shared space together. FAISS (inner-product) retrieves top-k evidence with condition-group filtering.
- Generation:** Top-k evidence injected into a grounded prompt; Llama-3.1-8B generates a Hinglish response. Zero-shot uses the same LLM, without evidence.
- Code Mixing Index (CMI):** Ratio of Hindi-origin tokens to total tokens; split into levels with low / medium / high tertiles for H2 analysis.
- Evaluation:** Concept-based scoring across 24 clinical categories with negation detection. **Factual Support** = output concepts found in evidence. **Hallucination** = output concepts absent from evidence.

## HYPOTHESES AND RESULTS

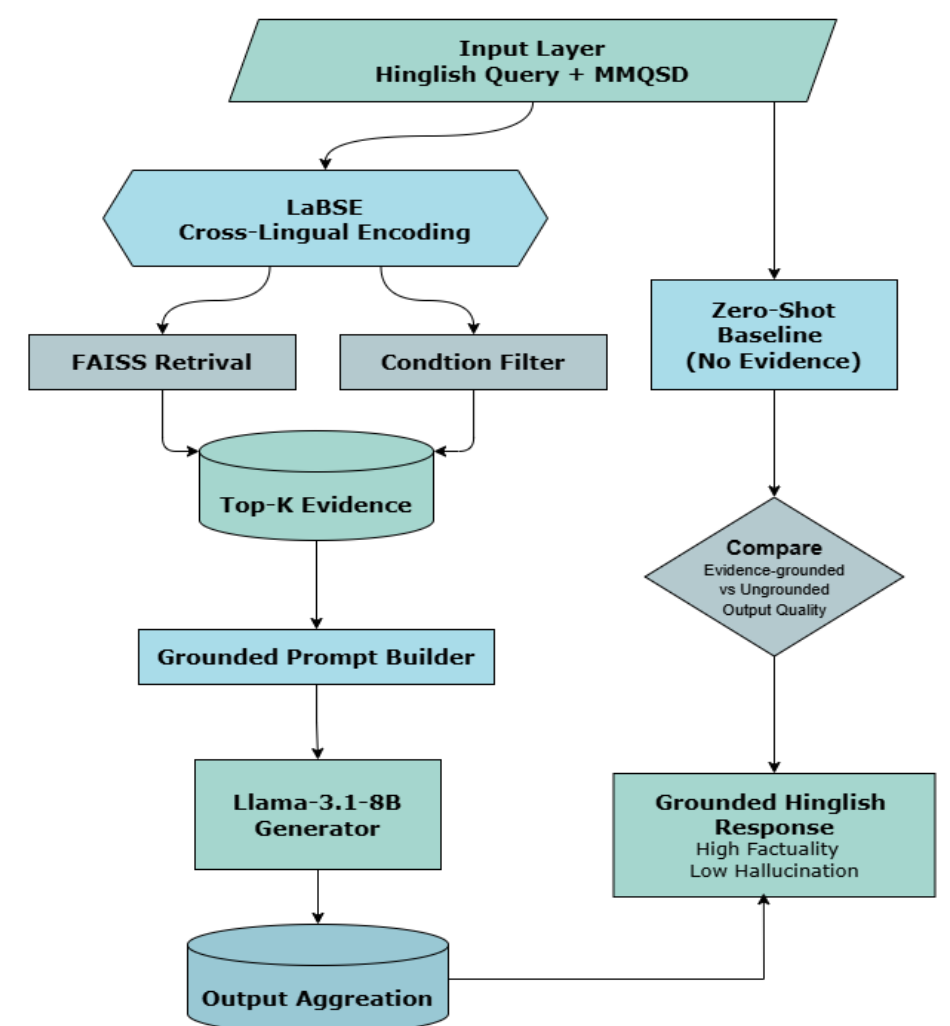
**H1:** Evidence-grounded RAG significantly outperforms zero-shot LLMs in factual consistency and hallucination reduction. (*wilcoxon signed-rank test, paired*)

| Metric          | Zero-Shot                               | Grounded | Change |
|-----------------|-----------------------------------------|----------|--------|
| Factual Support | 0.319                                   | 0.554    | +73.5% |
| Hallucination   | 0.500                                   | 0.280    | -44.0% |
| Decision        | Supported with significant improvement. |          |        |

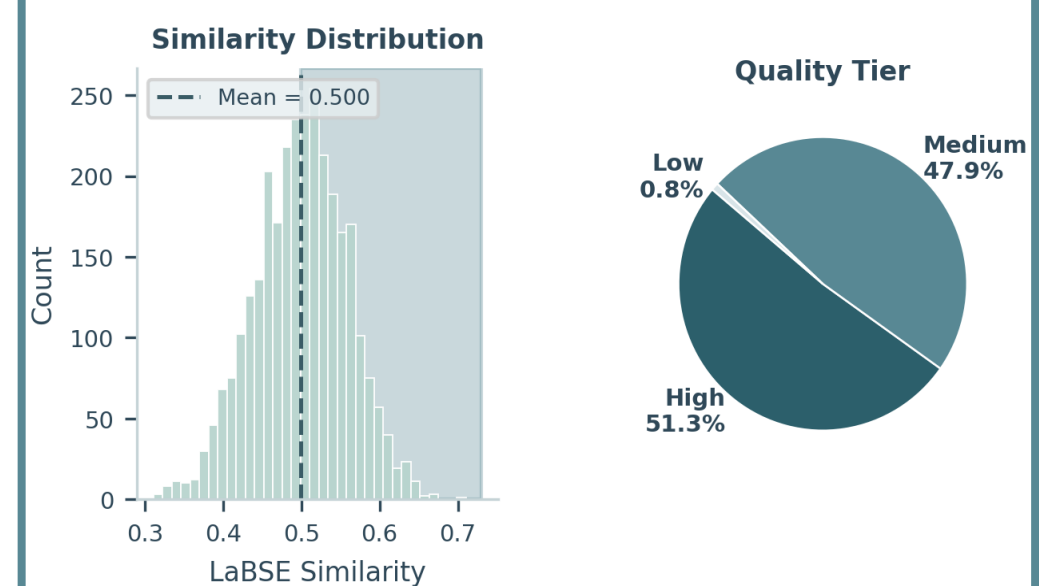
**H2:** Different code-mixing levels (Hinglish intensity) do not significantly degrade grounded model performance. (*kruskal-wallis h-test*)

| CMI Level | n                                    | Grounded Factual | Factual Gain | Hallucination Reduction |
|-----------|--------------------------------------|------------------|--------------|-------------------------|
| Low       | 385                                  | 0.554            | +0.202       | +0.206                  |
| Medium    | 384                                  | 0.544            | +0.241       | +0.208                  |
| High      | 396                                  | 0.563            | +0.260       | +0.245                  |
| Decision  | No significant degradation observed. |                  |              |                         |

## SYSTEM ARCHITECTURE WORKFLOW



## Retrieval Match (N = 3,015) · 99% Medium+High

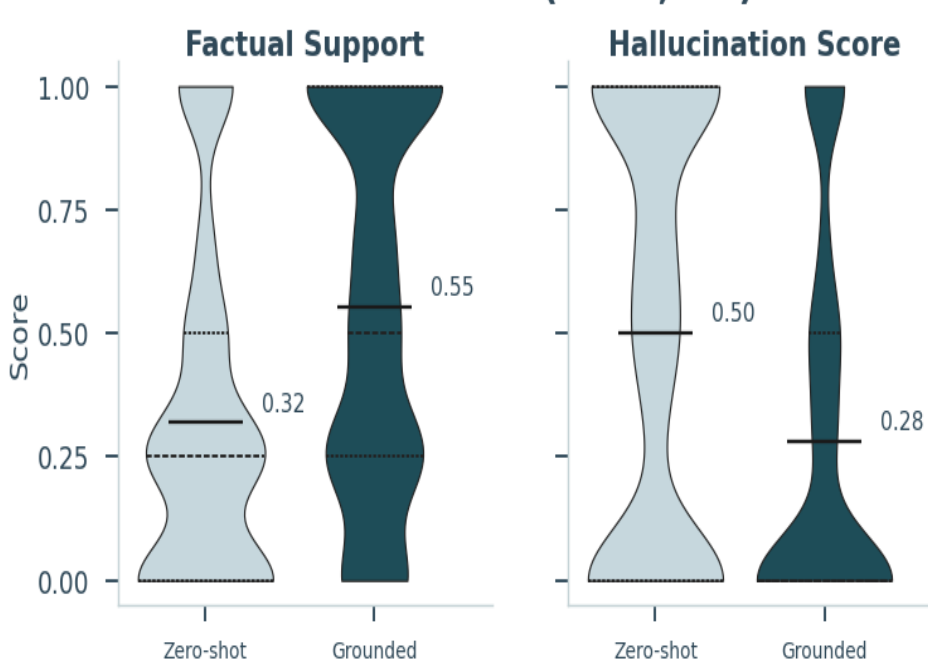


## CONCLUSION

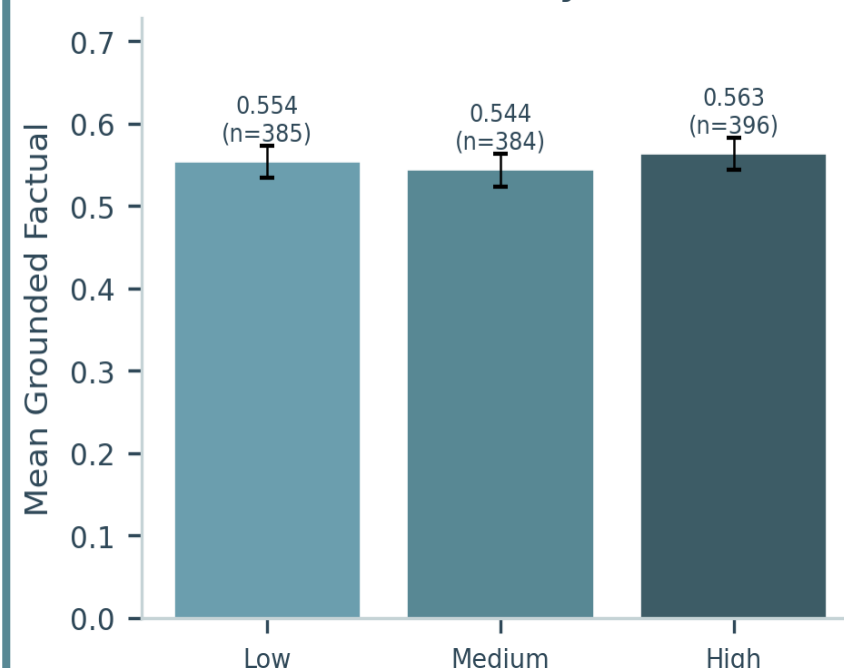
Evidence-grounded cross-lingual RAG yields safer, more fact-aligned Hinglish clinical answers than zero-shot generation alone, with stable performance across code-mixing, for supporting use in mixed Indian care settings. We adeptly implement retrieval → grounded prompting → scoring on the multi-specialty evidence, rather than imaging-only corpora.

**Future Work:** QLoRA fine-tuning, DPO-style alignment for hallucinations, and BioMedCLIP for image grounding.

## H1: Distributions (N = 1,165)



## H2: Grounded Factual By CMI (N = 1,165)



## MSC DS (B) – Group 5:

- Devika Jonjale: B045
- Saachi Shinde: B048
- Manjiri Apshinge: B061

## Mentors:

- Dr. Yogesh Naik
- Dr. Rajesh Maurya

## References:

- Ghosh et al. (2024) MedSumm: multimodal summarization of code-mixed Hindi-English clinical queries. ECIR 2024.
- Gumma et al. (2024) Health-Pariksha: assessing RAG models for multilingual health chatbots. arXiv.
- Ghosh et al. (2024) HealthAlignSumm: alignment-based multimodal summarization of code-mixed healthcare dialogues. Findings of EMNLP 2024.